

Clash Types & Priority Matrix

LUA BIM LABS Starter - Track 5 Quick Reference | Track completion card - Day 38

Clash Type Definitions

Clash Type	Definition	Example
Hard Clash	Two solid elements physically intersect - they	HVAC duct passing through a structural beam web
Soft / Clearance Clash	Two elements are within a defined minimum clearance distance - they do not touch but violate	Cable tray within 50mm of a sprinkler pipe, violating minimum separation
Duplicate	Two identical elements are stacked exactly the same position - usually a modeling or import error	Two pipe segments with the same path and elevation from different model imports
Workflow Clash (4D)	Two activities are scheduled for the same location at the same time - a time-space conflict, not a geometric one	HVAC installation and structural slab formwork both planned for Level 5 in the same week

Priority Level Matrix

Priority	Criteria	Required Action
Critical	Blocks structural work, prevents slab completion, or stops a safety-critical system from being installed	Must be resolved before any work in the affected zone proceeds - escalate immediately
High	Requires rerouting of a major system (main duct trunk, primary pipe run) - affects multiple connected elements	Resolve within current coordination cycle; assign to responsible trade with deadline
Medium	Requires a minor adjustment (elevation shift, small offset) - affects one or a few elements	Resolve before coordination drawing is issued for the zone
Low	Minor clearance violation with documented acceptance option; or affects only a non-critical element	Review and accept with documentation, or resolve at next available opportunity

Clash Grouping Rules

Before any coordination meeting, group raw clash results to reduce noise and focus effort:

- Group by Level: Separate clashes by floor so each trade addresses one zone at a time
- Group by Grid Bay: Cluster clashes within the same structural bay for efficient spatial review
- Group by Discipline Pair: Separate HVAC-vs-Structure from Electrical-vs-Plumbing - different trades, different meeting agenda items
- Filter by Status: Show only New and Active items - exclude Reviewed, Approved, and Resolved from the live discussion list
- Remove Duplicates: Many raw clashes are the same physical conflict reported from both element directions - review and consolidate before presenting

Coordination Meeting Agenda Template

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MEP COORDINATION MEETING - [Zone / Level] - [Date]

1. Attendance (5 min)
 - Record all attendees and represented trades
2. Previous actions review (10 min)
 - Confirm resolution of items from last meeting
 - Update status of Resolved items in tracker

3. Clash review - Critical items (15 min)
 - Screen-share Navisworks / clash report
 - Decision: Fix / Hold / Accept for each item
 - Assign owner and deadline
4. Clash review - High priority items (15 min)
 - Same process as Critical
5. New issues flagged by trades (10 min)
 - Any site or model issues not yet in the tracker
6. Actions summary (5 min)
 - Read back all assigned actions, owners, and due dates
7. Next meeting date confirmed

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